

Internship V

Torsional pendulum: Controller design for hanging and standing pendulums using the root locus curve

Control Engineering & Advanced Control Group – November 10, 2025

Building upon the existing speed control of the DC motor, this practical exercise investigates and controls the system with an attached pendulum. It is assumed that a functioning current and superimposed speed control cascade is implemented. The system input is therefore the target speed for the speed control. The system output is the angle of the pendulum. The goal is to design and implement an angle control for both the suspended and the stationary pendulum. We will use the root locus method for the controller design.

Learning objectives

- Linearization of a nonlinear mechanical system around given operating points
- Derivation and interpretation of transfer functions
- Design of a PI controller using the root locus
- Implementation and testing of the rule on the real system including enable/reset logic

1 Differential equation

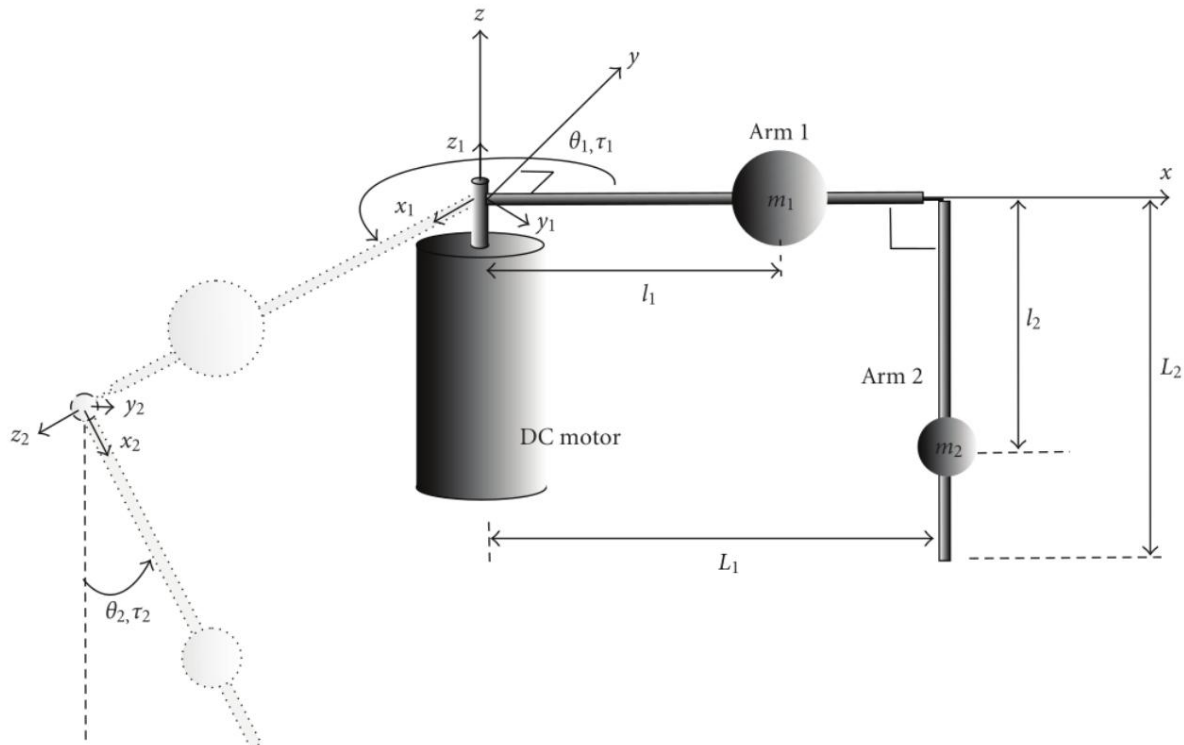


Figure 1: Sketch of torsional pendulum from [1].

The complete derivation of the nonlinear equations of motion for the torsional pendulum is described in detail in [1] (2 mechanical DOF → 2 second-order nonlinear differential equations). Neglecting Coriolis forces, the nonlinear differential equation for the pendulum is:

$$J_P + m_2 L^2 \ddot{\gamma}_2 + b_2 \dot{\gamma}_2 + m_2 g l_2 \sin \gamma_2 = \ddot{\gamma}_1 m_2 L l_2 \cos \gamma_2$$

To be consistent with previous practical exercises, we will continue to use the designations γ_1 for the angle of the motor and γ_2 for the angle of the pendulum instead of $\dot{\gamma}_1$ and $\dot{\gamma}_2$ for the angles.

$$J_P + m_2 L^2 \ddot{\gamma}_2 + b_2 \dot{\gamma}_2 + m_2 g l_2 \sin \gamma_2 = \ddot{\gamma}_1 m_2 L l_2 \cos \gamma_2$$

It should be noted that:

- The zero angle of the pendulum is given when the pendulum is suspended.
- The direction of rotation of the rotary axes is chosen to be mathematically positive (right-hand rule).

Size	Value
Motor arm length L_1 , moment of inertia of the pendulum relative to COG J_P	0.0855 m
$6.063 \times 10^{-5} \text{ kg m}^2$, mass of the pendulum m_2	
0.028 kg, distance to center of gravity l_2	
0.0805 m, coefficient of friction b_2 , acceleration due to gravity g	
	$4.4 \times 10^{-5} \text{ N m/(rad s)}$
	9.81 m/s^2

Table 1: Parameters of the torsional pendulum

Tasks

1. Sketch the forces and moments acting on the pendulum.
2. What is the system input and what is the system output?

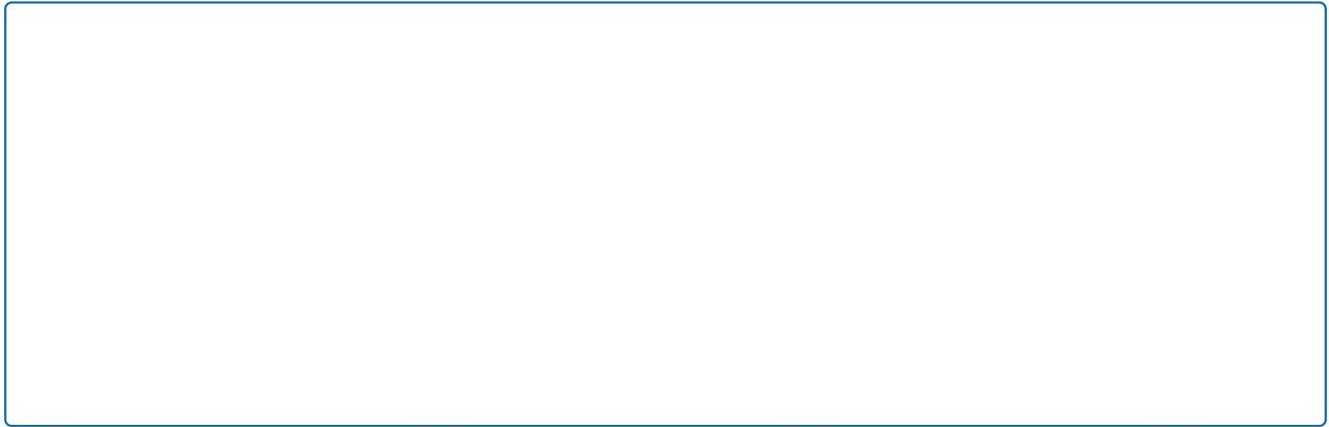
2 pendulums hanging

In the first part, the pendulum is examined in its limitingly stable equilibrium position (neglecting friction), i.e. , *in its suspended state* . The aim is to analyze the system behavior around this operating point and to design a suitable controller. For this purpose, the nonlinear differential equation is first linearized and transformed into a transfer function. Subsequently, the stability and static behavior are evaluated before the controller is designed in the next step, based on the root locus method .

Systems analysis

Tasks

1. Linearize the differential equation symbolically around the operating point of the suspended pendulum ($\ddot{\gamma}_2, 0 = 0$) . Use the approximation for small angles: $\sin \ddot{\gamma}_2 \approx \ddot{\gamma}_2$ and $\cos \ddot{\gamma}_2 \approx 1$.
2. Determine the transfer function of the system input *angular acceleration of the motor* $\ddot{\gamma}_1(s)$ to system output *pendulum angle* $\ddot{\gamma}_2(s)$.
3. We are using a torsional pendulum with a subordinate speed control. Therefore, determine the transfer function from the system input *angular velocity of the motor* $\dot{\gamma}_1(s) = \ddot{\gamma}_1(s)$ to the system output *pendulum angle* $\ddot{\gamma}_2(s)$. Also, change the sign of the controlled system so that a positive input results in a positive output change. Note that you must also take this sign into account in the real system.
4. Neglect friction ($b_2 = 0$) and calculate the transfer function using numerical values.
5. Determine the poles and zeros of the transfer function.
6. What type of system is it in terms of static behavior and stability?



Controller design

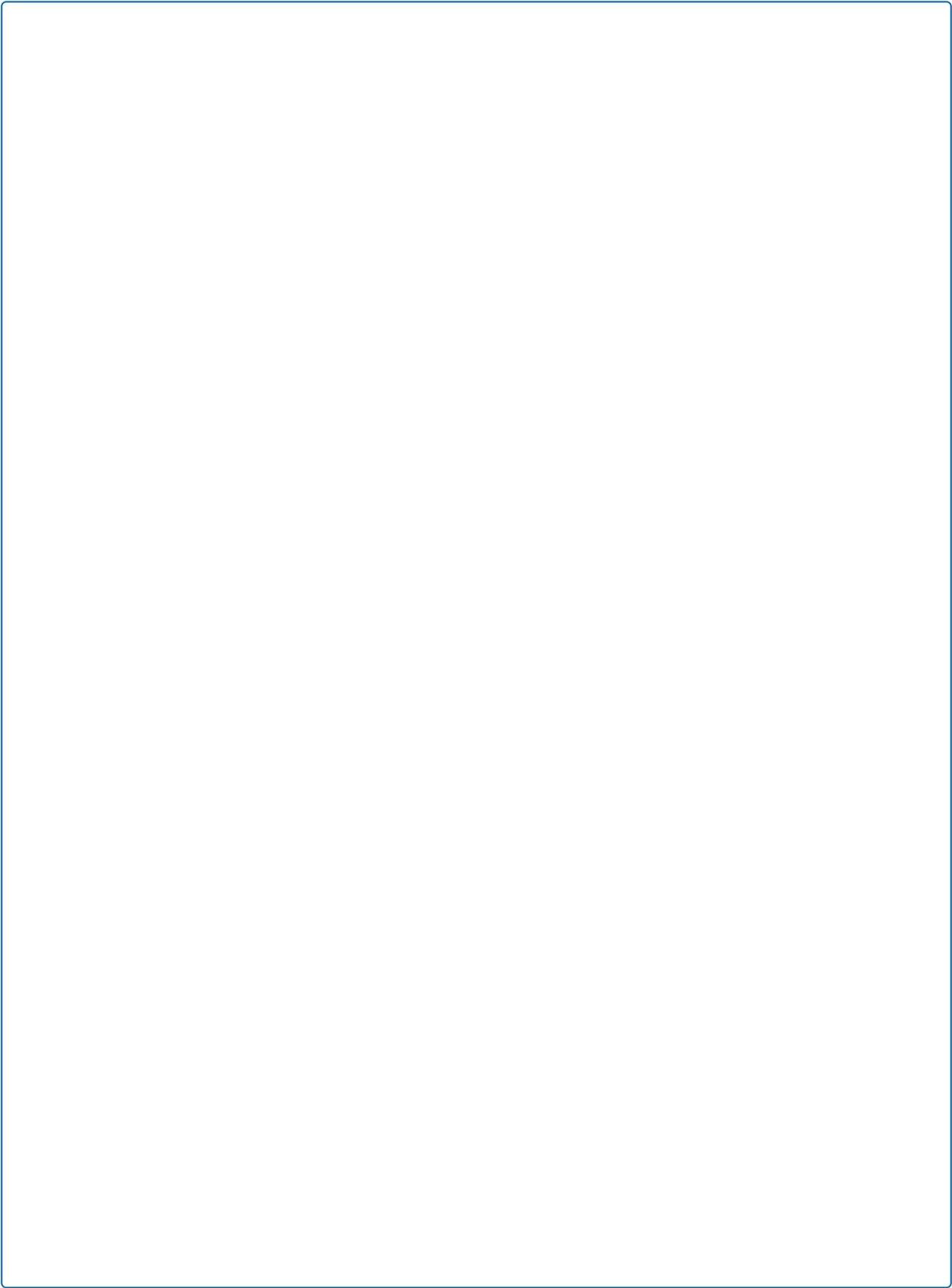
The controller to be designed is intended to dampen the suspended pendulum. Using the root locus method, the movement of the poles of the closed-loop control system in the complex plane is investigated as a function of the gain. Based on this, the reset time T_i is first specified, and then the required gain k_p is determined, such that the closed-loop control system has a double pole. The dynamics of the speed control can be neglected here, as they are significantly faster than the dynamics of the pendulum.

We choose a PI controller of the following type as the controller structure:

$$C(s) = k_p \frac{T_i s + 1}{T_i s}$$

Tasks

1. Determine the reset time T_i , so that a zero is created at $z = \tilde{y}_9$.
2. Determine the transfer function of the open-*loop* control system $L(s)$ using numerical values. If possible, cancel out poles and zeros that do not lie in the right half-plane.
3. Normalize the open-loop transfer function according to $k\tilde{L}(s)$.
4. Sketch the root locus curve, proceeding as follows:
 - 4.a) Sketch the poles and zeros of $\tilde{L}(s)$ in the complex plane.
 - 4.b) Determine the polar excess and the intersection point of the asymptotes s_A .
 - 4.c) Consider which parts of the real axis belong to the root locus.
 - 4.d) Determine the candidates for the branching points (*breakaway and break-in points*). Decide which of the candidates are actually branching points and enter them in the sketch.
 - 4.e) Complete the sketch of the root locus curve.
 - 4.f) Using the *magnitude condition*, determine the gain k such that a double real pole is created. Then determine the controller gain k_p .



Implementation and testing of the regulation

In the received Simulink template, the control system is implemented such that both the PI controllers themselves and the actual motor driver are only active when the angle is within a predefined range (see Simulink model). The logic block is fundamentally designed so that the controllers are reinitialized upon activation (integrator reset). Nevertheless, the logic block is implemented in such a way that the control system and the power electronics are only activated once. If the pendulum moves outside the range, the control system is deactivated and the motor is switched off. To reinitialize the system, the entire *Simulink model* must be stopped and restarted. Superimpose the PI angle control onto the existing speed control, including the enable/reset logic, and take into account the negative sign at the target speed input of the speed control.

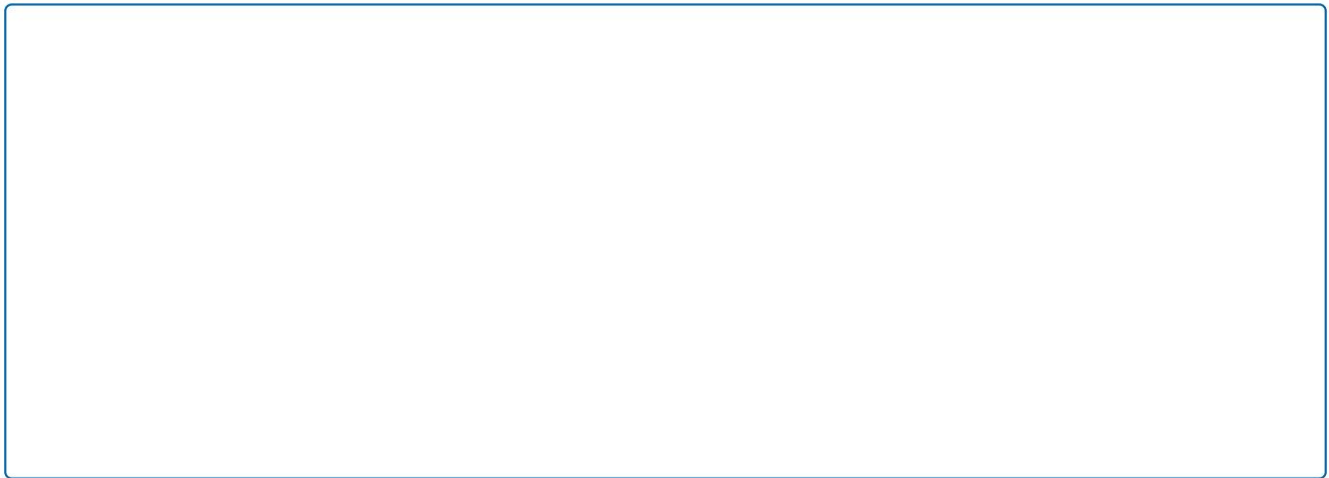
3 pendulums standing

In the second part of the practical exercise, the pendulum is examined in its unstable equilibrium position, i.e., when *stationary*. Again, we linearize the nonlinear differential equation around this operating point and transform it into a transfer function. Equivalent to the first part, the stability and static behavior are evaluated before, in the next step, the stabilizing controller is designed based on the root locus method.

Systems analysis

Tasks

1. Linearize the differential equation of a pendulum standing around the operating point ($\ddot{\gamma}_2, 0 = \ddot{\gamma}$) symbolically. Use the approximation for small angles: $\sin(\ddot{\gamma}_2 - \ddot{\gamma}) \approx \ddot{\gamma}_2 - \ddot{\gamma}$ and $\cos(\ddot{\gamma}_2 - \ddot{\gamma}) \approx 1$.
2. Determine the transfer function from system input *angular velocity of the motor* $\dot{\gamma}_1(s) = \ddot{\gamma}_1(s)$ to system output *pendulum angle* $\ddot{\gamma}_2(s)$. The sign of the controlled system does not need to be changed here. The zero position of the pendulum is given around the operating point when the pendulum is stationary.
3. Neglect friction ($b_2 = 0$) and calculate the transfer function using numerical values.
4. Determine the poles and zeros of the transfer function.
5. What type of system is it in terms of static behavior and stability?

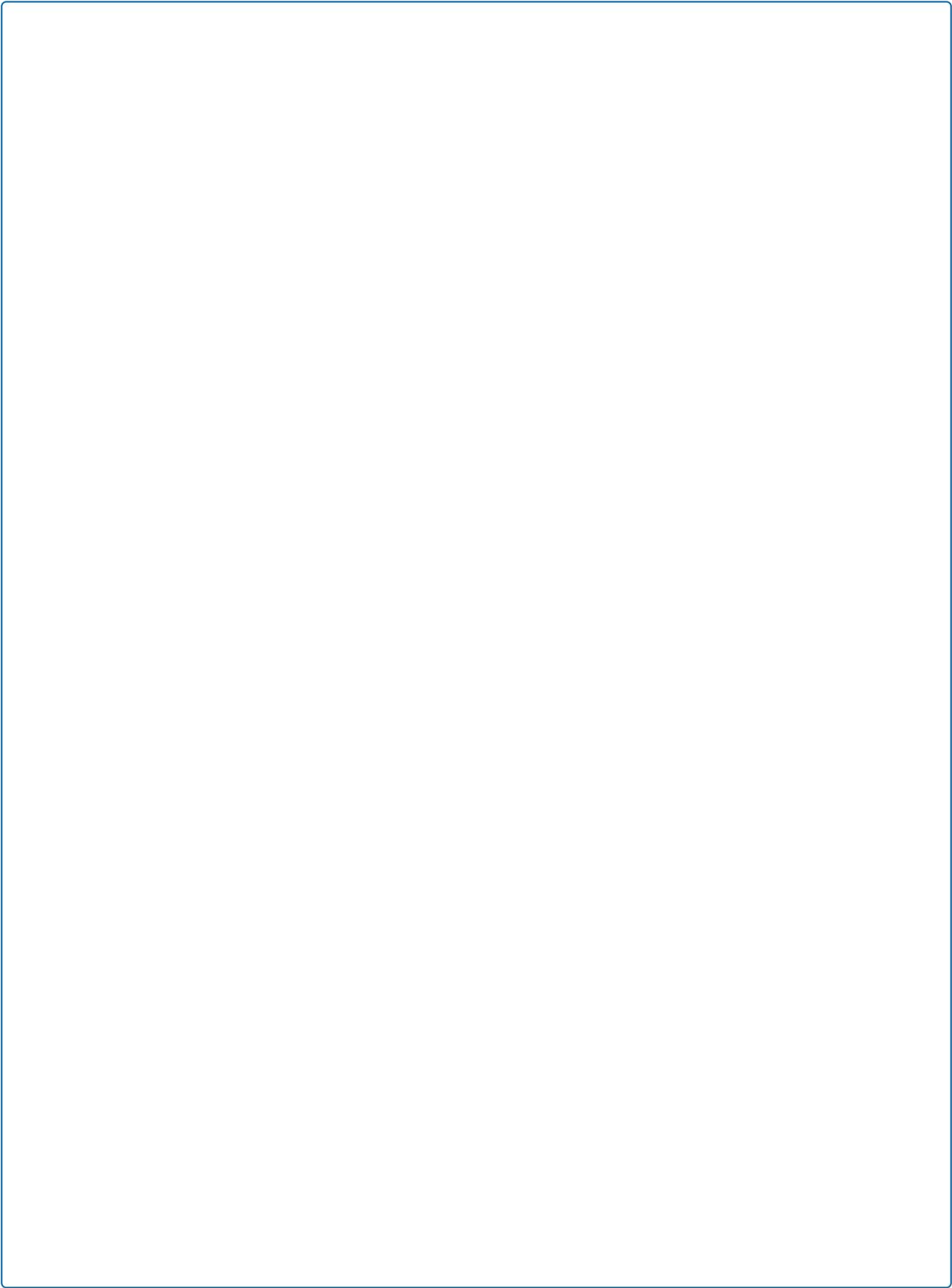


Controller design

The controller to be designed should now stabilize the stationary pendulum. The goal is therefore to shift the unstable pole of the controlled system into the left half-plane. As with the hanging pendulum, we use a PI controller and design it using the root locus method. The zero of the PI controller is again predetermined, so we can concentrate solely on constructing the root locus and determining the gain.

Tasks

1. Determine the reset time T_i , so that a zero is created at $z = -18$.
2. Determine the transfer function of the open- *loop* control system $L(s)$ using numerical values. If possible, cancel out poles and zeros that do not lie in the right half-plane.
3. Normalize the open-loop transfer function according to $k\tilde{L}(s)$.
4. Sketch the root locus curve, proceeding as follows:
 - 4.a) Sketch the poles and zeros of $\tilde{L}(s)$ in the complex plane.
 - 4.b) Determine the polar excess and the intersection point of the asymptotes s_A .
 - 4.c) Consider which parts of the real axis belong to the root locus.
 - 4.d) Determine the candidates for the branching points (*breakaway* and *break-in points*). Decide which of the candidates are actually branching points and enter them in the sketch.
 - 4.e) Complete the sketch of the root locus curve.
 - 4.f) Determine the gain k using the *magnitude condition* .
The root locus curve shows that we have two possibilities for parameterizing a double pole of the closed-loop control system. Determine both gains and then the corresponding controller gains k_p .
 - 4.g) Now consider how the gain k can be chosen so that we design a system in which one pole lies at the highest point of the root locus and one pole lies at the lowest point of the root locus. Then determine the corresponding controller gain k_p and implement this controller.



Implementation and testing of the regulation

Implement the angle control for the stationary pendulum analogously to the suspended pendulum. Note that the sign at the target speed input of the speed control must be positive here. Furthermore, the pendulum's zero position around the operating point is given when the pendulum is stationary. Consequently, you must subtract $\dot{y}$ from the measured angle $\ddot{y}_2$ to obtain the operating point under consideration.

literature

- [1] BS Cazzolato and Z. Prime, "On the Dynamics of the Furuta Pendulum," *Journal of Control Science and Engineering*, vol. 2011, Article ID 528341, 8 pages. doi: [10.1155/2011/528341](https://doi.org/10.1155/2011/528341).